

PROJECT DEVELOPMENT PHASE

DATE	04/10/2026
TEAM ID	SWTID-2026-8904
PROJECT NAME	AI Study Buddy

MAXIMUM MARKS: 5 Marks

Project Overview

Project Name: AI Study Buddy
Project Description: An AI-powered study assistant that helps students understand topics, organize study materials, generate quizzes, and track learning progress. The system uses Google Gemini AI for explanations and study assistance.
Project Version: 1.0
Testing Period: 29/09/2026 to 04/10/2026

Testing Scope

Area	Testing Scope
Authentication	Student registration, login, JWT validation and protected routes.
Study Materials	Create, store, update, retrieve and search study notes.
AI Assistant	Topic explanations, question answering and study suggestions using Gemini API.
Quiz	Generate quiz questions, submit answers and record results.
Progress	Store and retrieve study history, quiz scores and completed topics.
Database	Verify users, notes, quizzes and progress data in MongoDB.
Integration	Verify frontend and backend API communication.

Test Cases

ID	Scenario	Test Steps	Expected Result	Status
TC-001	Registration	Enter name, email, password and register.	Student account created successfully.	Pending
TC-002	Login	Enter registered credentials.	Student logs in successfully.	Pending
TC-003	AI Explanation	Enter a study topic and submit.	Clear AI-generated explanation is displayed.	Pending
TC-004	Quiz Generation	Request a quiz for a topic.	Relevant quiz questions are generated.	Pending
TC-005	Notes	Create and retrieve a study note.	Note is saved and displayed correctly.	Pending
TC-006	Progress	Complete a quiz and open progress page.	Score/history is displayed correctly.	Pending

Bug Tracking

Bug ID	Description	Severity	Status	Feedback
BG-001	AI explanation not generated, if encountered.	Medium	To be tested	Verify Gemini API configuration.
BG-002	Login failure, if encountered.	Medium	To be tested	Verify JWT and credentials.
BG-003	Study note not saved, if encountered.	High	To be tested	Verify MongoDB connection.

Sign-off: Tester Name: [Your Name] Date: 04/10/2026 Signature: [Tester's Signature]